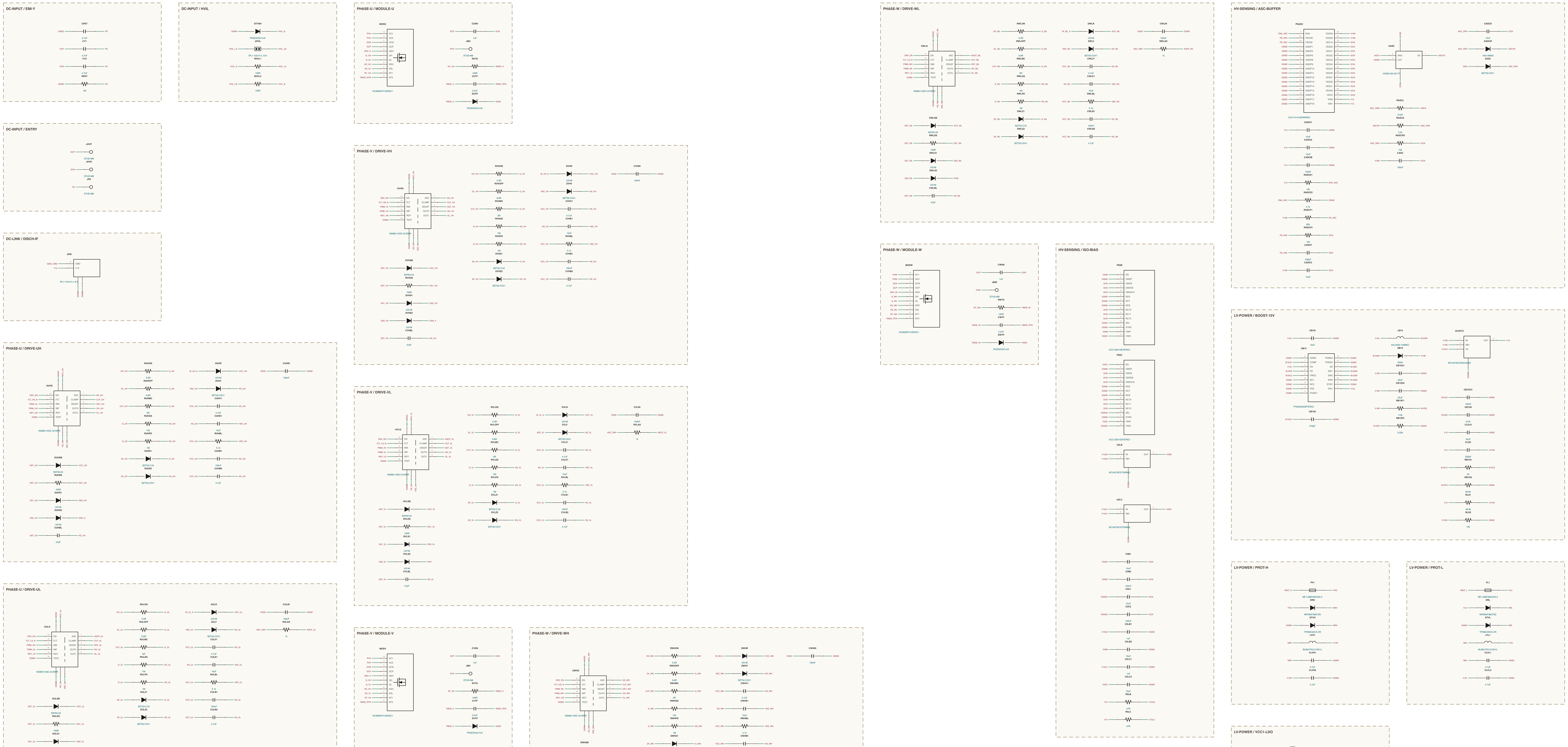
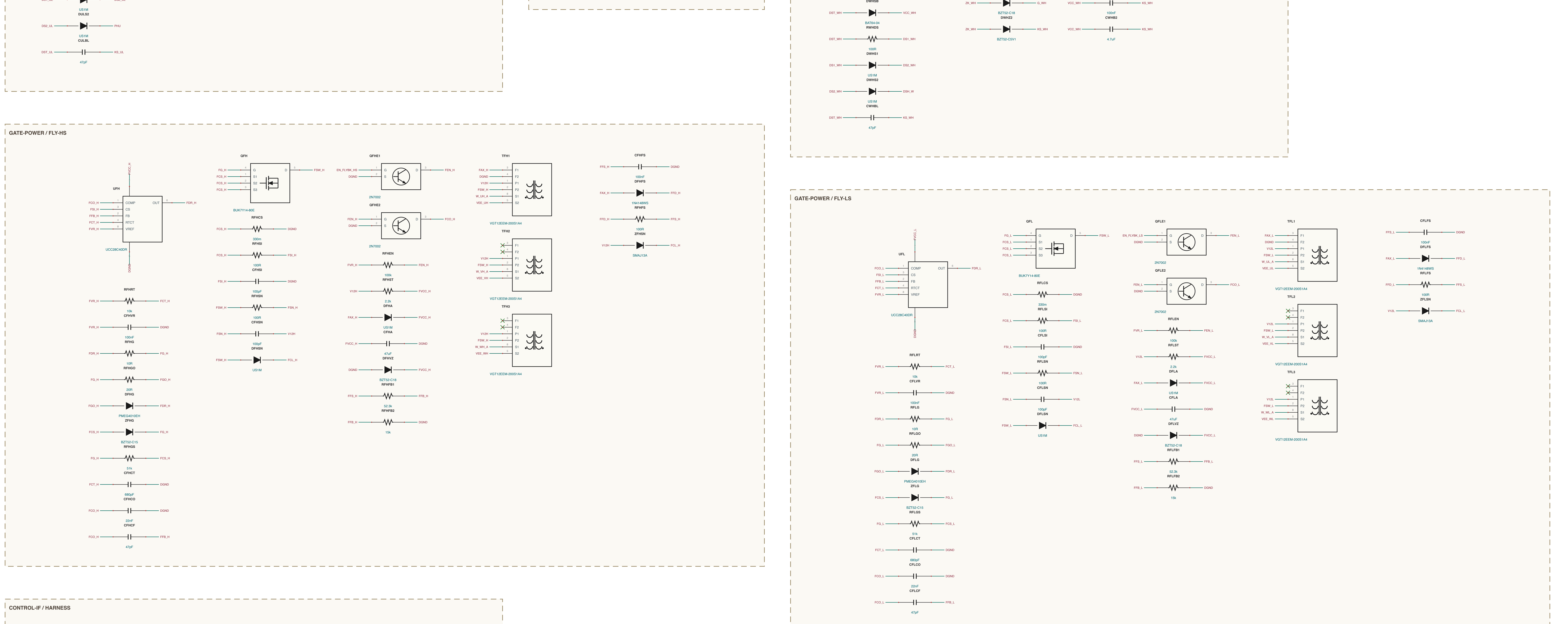
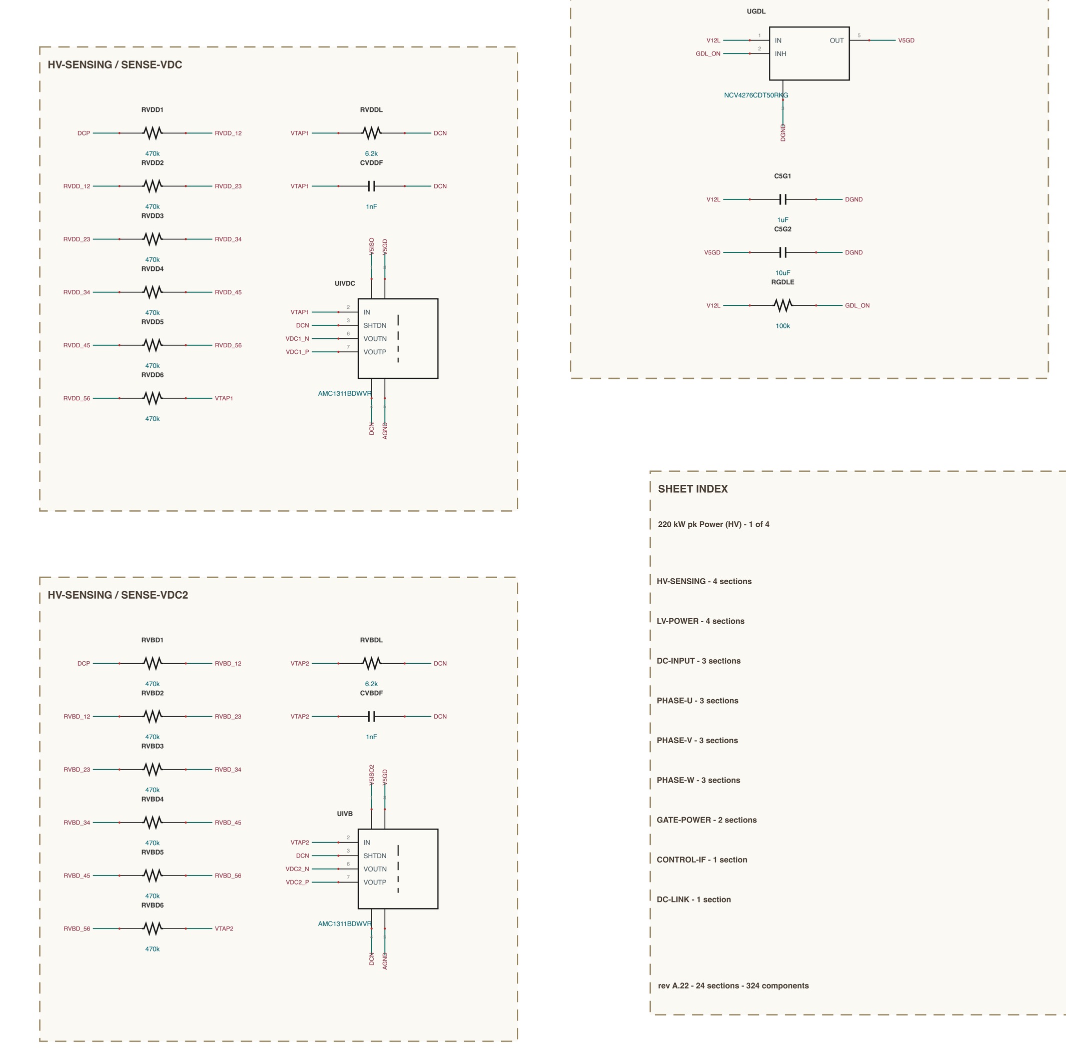
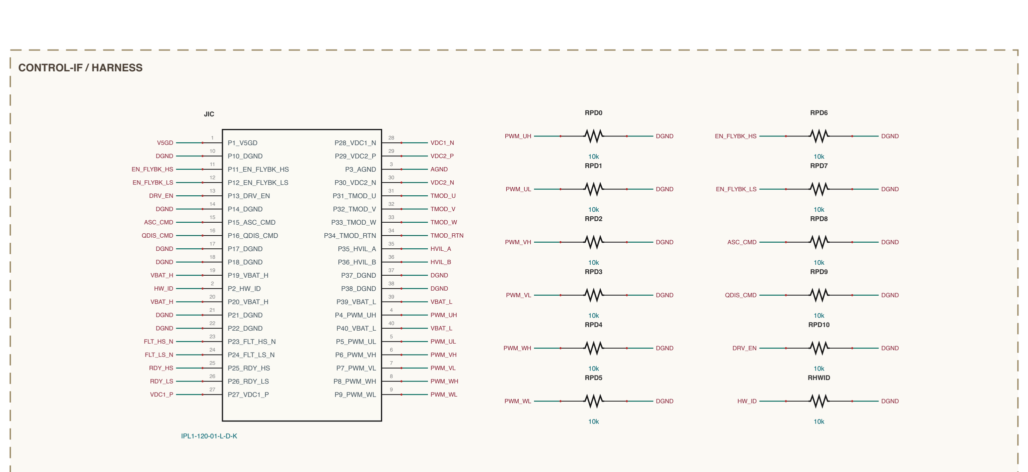








NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVDC_VINP

R<stem>_M midpoint of a series pair e.g. RBLA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RVDD_12

all others are explicit design nets

same name on another sheet = same net, joined ONLY at the

named connector/stud/tab (see INTERBOARD LINKS, design basis)

SHEET INDEX

220 kW pk Power (HV) - 1 of 4

HV-SENSING - 4 sections

LV-POWER - 4 sections

DC-INPUT - 3 sections

PHASE-U - 3 sections

PHASE-V - 3 sections

PHASE-W - 3 sections

GATE-POWER - 2 sections

CONTROL-IF - 1 section

DC-LINK - 1 section

rev A.22 - 24 sections - 324 components

ASIL-D SAFETY CONCEPT + DC-LINK DISCHARGE VERIFICATION

safe state = 3-phase-open (ASC above overspeed); discharge vs ECE R100 / ISO 6469

1 lockstep S32K396 (ASIL-D core) runs torque path

2 FS26 Q&A watchdog + monitors -> FS0B (no SW)

3 FS0B&MCU_EN&RDY&FLT-latch -> DRV_EN = 3-ph open

4 FS1B strap -> LS ASC latch (HS strapped OFF)

5 flyback EN = MCU OR FS26-GPIO1 (ASC gate power)

6 3x hall + sum(I)=0 plausibility (torque path)

7 2x iso VDC senses >5% = fault, VOFS monitored

8 DESAT per switch (SiC <=1.9 us detect), latching

9 HVIL open = VDDIO/2 signature -> torque off (FW)

--- DISCHARGE ---

PASSIVE 12x 22k 2512-2W (6s x 2p = 66k): 56 s to 60 V

ACTIVE QDIS + 4x 470R 10W = 1.88k: 1.6 s to 60 V

E = 116 J @850 V - 32 J/res worst; 4XX: 220R/15k

VERIFY: MCU fires QDIS, watches BOTH VDC channels

RULE: never energize w/o the DISCHARGE BOARD fitted

Traction Inverter 220 kW — Power board (3x EconoDUAL 3 SiC)

Traction Inverter 220 kW pk - Power (HV), sheet 1 of 4

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